

Areg Hovumyan

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Education

California State Polytechnic University, Pomona
Bachelor of Science in Computer Engineering — GPA: 3.5/4.0

Pomona, CA
Expected December 2027

Experience

Electrical/Embedded Systems Member: L'SPACE (Internship/Academy)

Remote

NASA's student acceleration program - Robotic Lunar Rover Mission

May 2026 – Present

- Present the Communication and Data Handling to NASA engineers for a possible mission consideration on the moon.
- Manage to stay within the schedule, cost, and mass limitations (under 250M, 170Kg, no sunlight for 14 days and more).
- Implement Computer Vision on the rover to explore permanently shadowed craters on the moon.
- Design the whole rover's Communication and Data Handling(C&DH) architecture by calculating the best-fitting battery, figuring out the communication protocols, and choosing the flight software by studying NASA's previous missions.
- Establish (on concept) the communication between the Earth and the Moon, autonomous mission completion in case of redundancy, and learn F* to fit the mission under the mission needs.

Software Engineer: Autonomous Vehicle Lab (Research)

Pomona, CA

University Research Lab

April 2026 – Present

- Create an interceptor drone, running object detection on a Jetson, establishing autonomous flight through a ready-made flight controller, and fitting under the budget of \$2000.
- Build a closed-loop stepper controller (CL57T) on a microcontroller with CAN telemetry over a TJA1051 transceiver to control a steering wheel through a computer, controller, or a joystick.
- Engineer CI/CD pipeline in GitHub Actions for the team to catch regressions before merge, and create unit tests to improve system reliability.
- Resolve priority-inversion bug by introducing mutex priority inheritance, restoring deterministic timing.
- Utilize AI agents to interface an MPU IMU over I2C on a Nucleo to stream 9-DOF telemetry.

Projects

Software Engineering Team Lead: Autonomous Hexacopter (University Club)

Pomona, CA

Lockheed Martin sponsored project

June 2025 – Present

- Apply pair programming, test-first development, unit testing, create containers for a ROS2 environment with Docker for a team of 8 student-engineers to collaborate, and create CI/CD pipelines to ship features with fewer defects.
- Currently integrating the hexacopter into Anduril's Lattice via a ROS2 bridge node that fuses MAVLink flight state with onboard perception, georeferencing pixel detections to WGS-84 ground coordinates and publishing them as C2 entities on the flight control loop.
- Collaborate with the Electrical team to not overwhelm the processor with object detection and image capturing processes, and work with the Payload team to drop payloads mid-flight using servos.
- Present mapping and Computer vision improvement, image processing achievement, and correct GPU/CPU utilization to Lockheed Martin engineers during the Preliminary Design Review (PDR).
- Create ROS2 nodes that handle the whole mission, including object detection (0.99 mAP@0.5), area mapping, 4k image processing (under 0.7s each), and payload release over a target.

Embedded Systems Member: Autonomous Ground Vehicle (University Club)

Pomona, CA

Northrop Grumman Collaboration Project

February 2026 – June 2026

- Present telemetry connection between a drone, rover, and a ground station workflow to Northrop Grumman engineers during a Preliminary Design Review (PDR).
- Implement Feature-driven development and the lean approach to release new code every week and meet all the requirements before any demonstrations or meetings.
- Run an authorized hardware review of a DS2431-class authenticator with a Bus Pirate v4 and logic analyzer, confirming a family-code mismatch and identifying clone silicon.
- Collaborate with the firmware team to develop bare-metal firmware for a pre-made microcontroller in C, directly manipulating registers (GPIO, timers, interrupts, etc.).

Skills

Languages and Tools: C, C++, Python, Rust, Java, VHDL, ARM Assembly, Bash, SQL, bare-metal, UART, SPI, ROS2 (Humble/Iron), Gazebo, YOLO, PyTorch, computer vision, machine learning, Linux, Git, GitHub Actions, Docker, MongoDB, Javascript, Typescript, Node.

Coursework: Microcontrollers & Embedded Systems, Circuit Analysis, Signals & Systems, Data Structures & Algorithms.

Languages: English, Russian, Armenian